

Analysis group meeting

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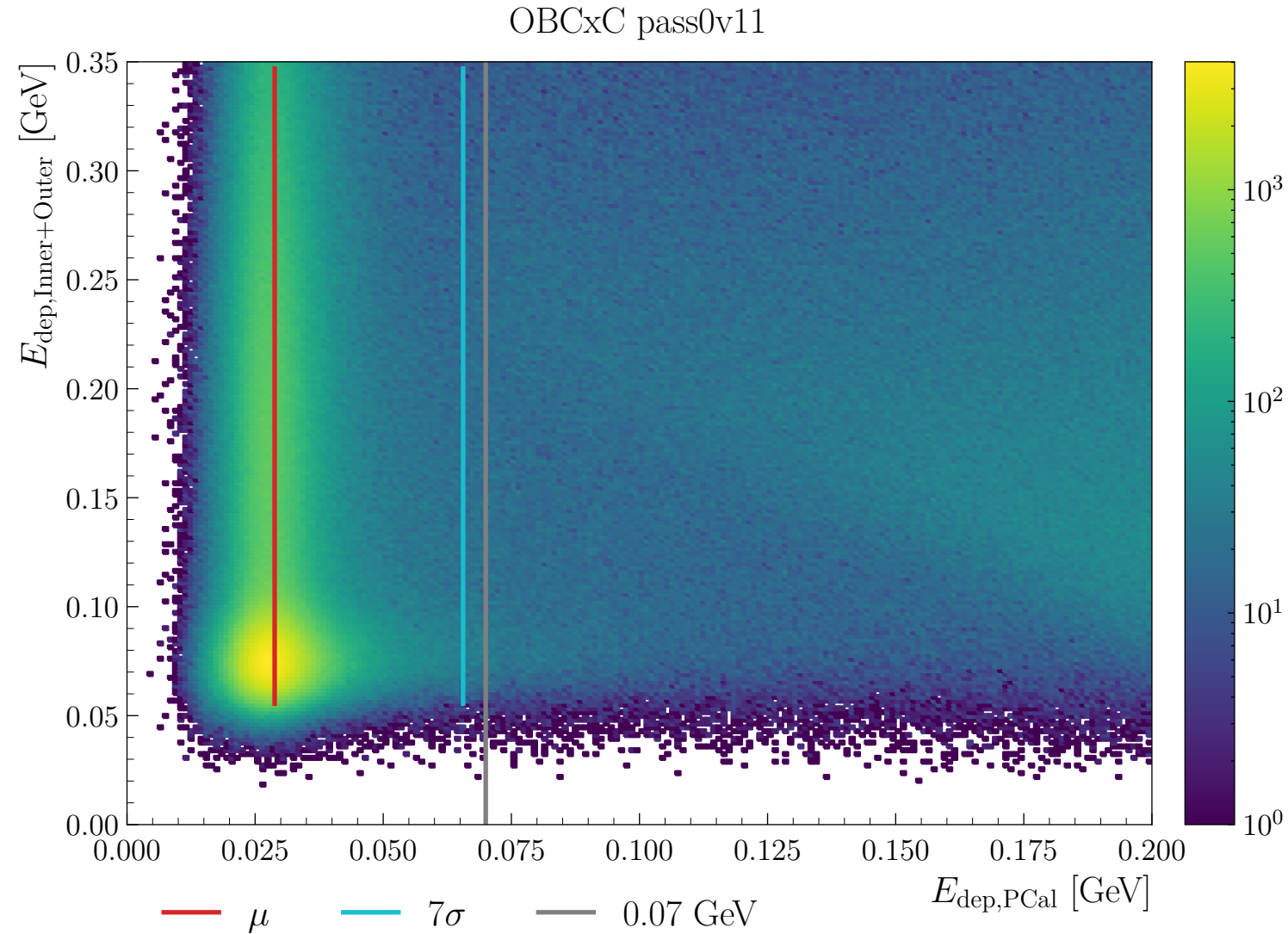
06/02/25

RG-D: Fiducial cuts

- To select a good electron, we use $\text{pid}=11$, $\text{status}<0$ and $|\text{chi2pid}| < 5$
- In the common analysis note for RG-A and RG-M, they improve these cuts:
 - PCAL minimum energy deposition cut: $E_{\text{PCal}} > 0.07 \text{ GeV}$
 - Sampling fraction vs momentum cut: $\mu \pm 3\sigma$
 - Inner calorimeter and the PCal SF cut: $\frac{E_{\text{dep,Inner}}}{p} < 0.2 - \frac{E_{\text{dep,PCal}}}{p}$
 - v and w cuts on the PCal: 9 cm
- DC fiducials cuts using the edge variable

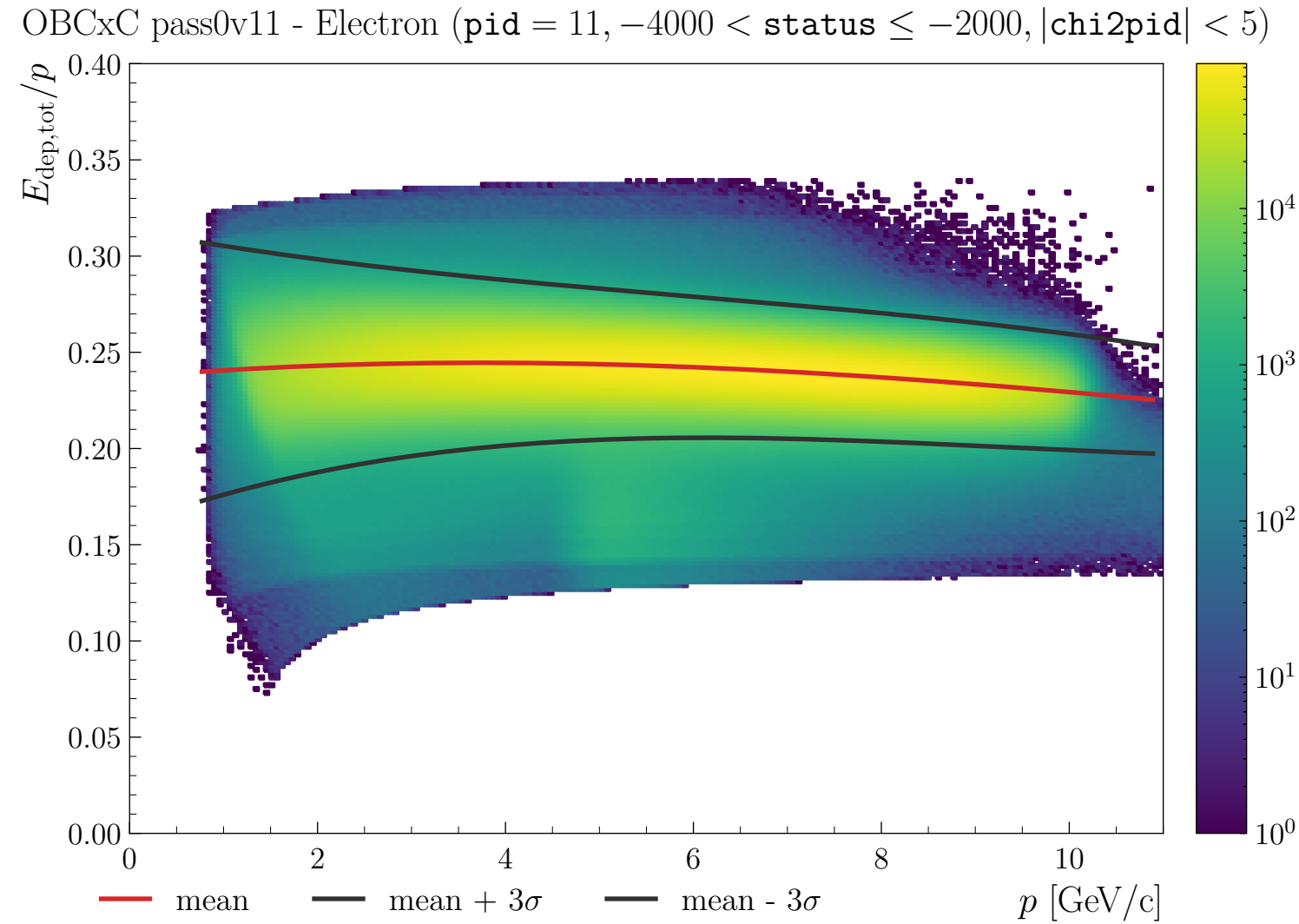
RG-D: PCAL minimum energy deposition cut

- Slice the 2D distribution along the y axis
- Fit the 1D distribution with gaussian
- Extract the mean and sigma, fit it with a constant
- A cut at 0.07 GeV is like a 7sigma cut



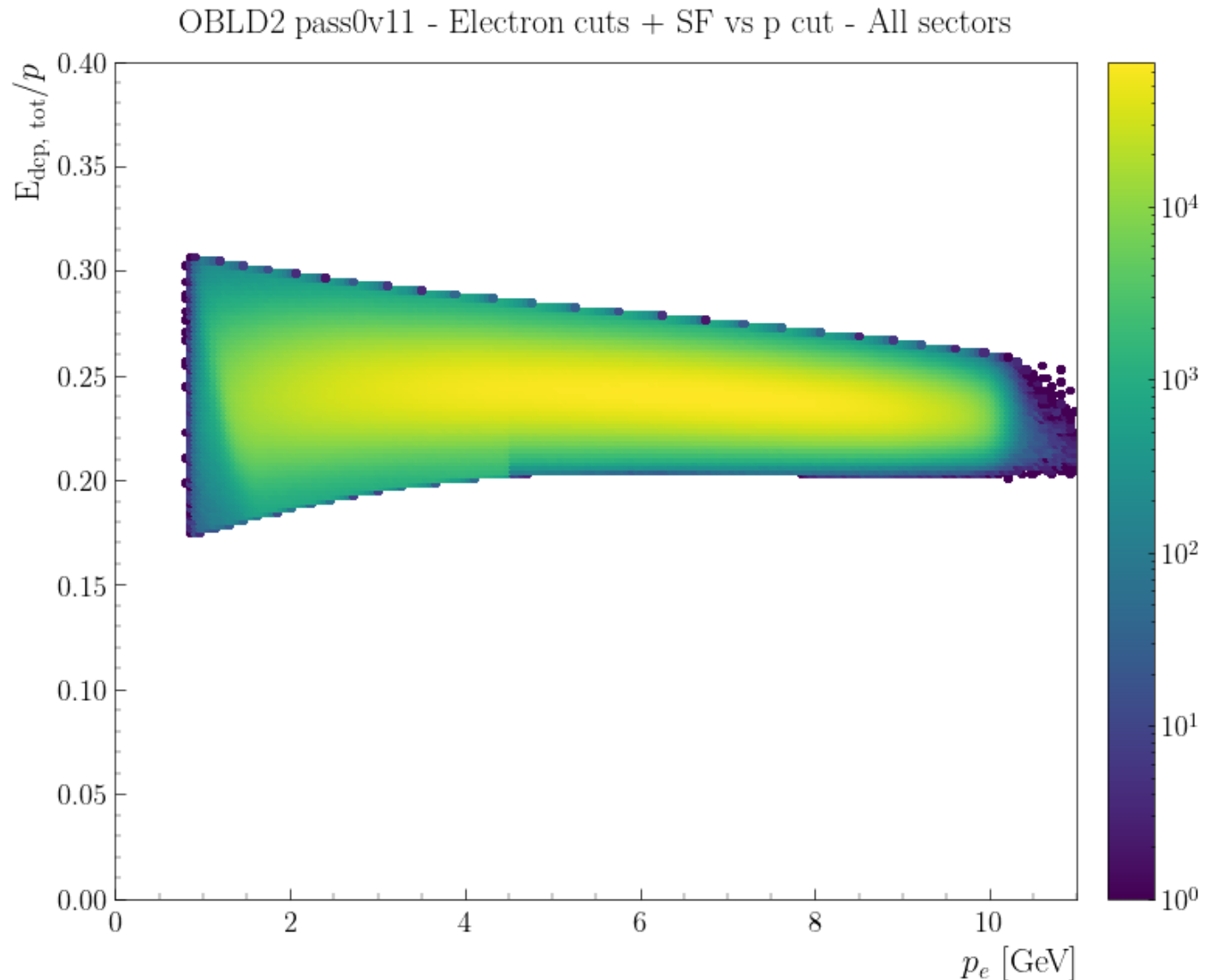
RG-D: Sampling fraction vs momentum cut

- Slice the 2D distribution along the x axis
- Fit the 1D distribution with gaussian
- Extract the mean and sigma, fit with a 3rd order polynomial function
- Cut at $\mu \pm 3\sigma$



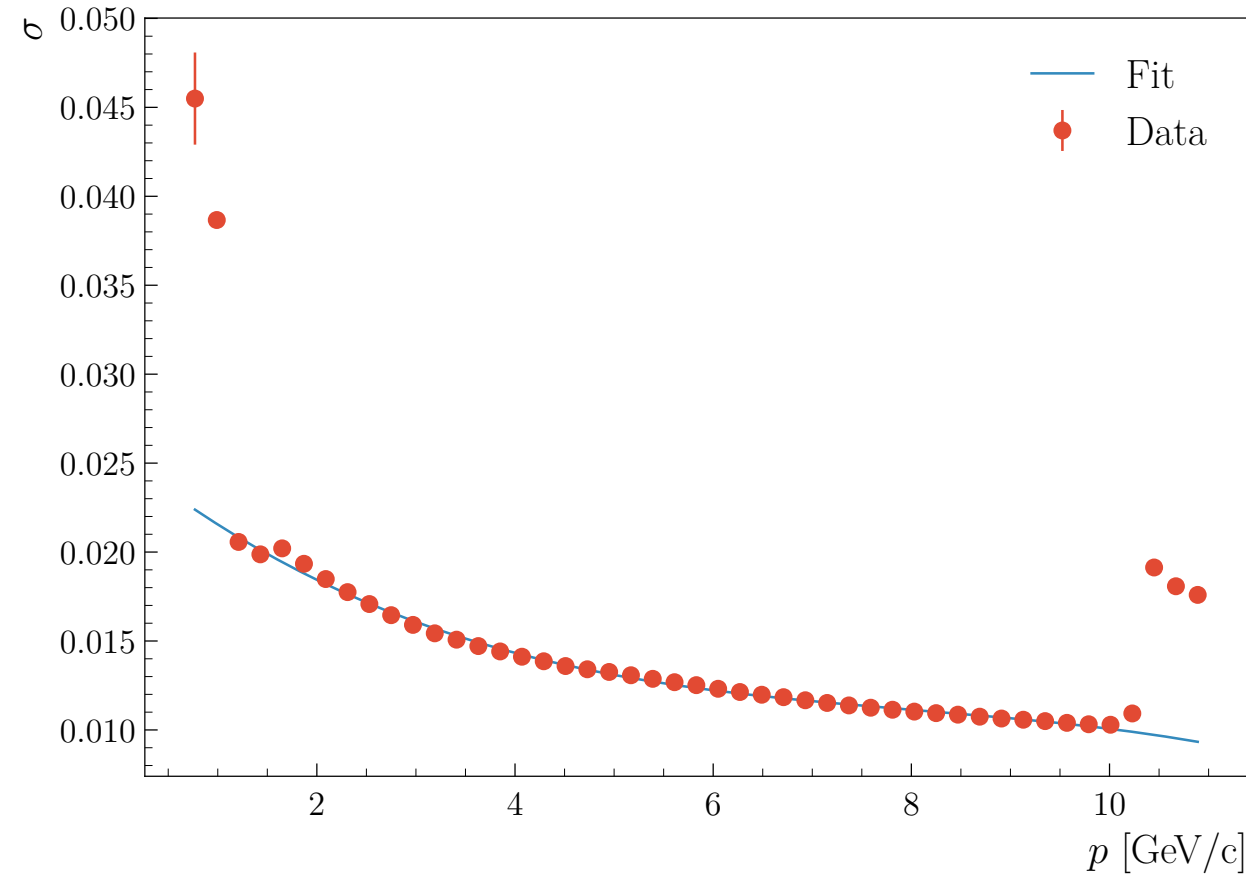
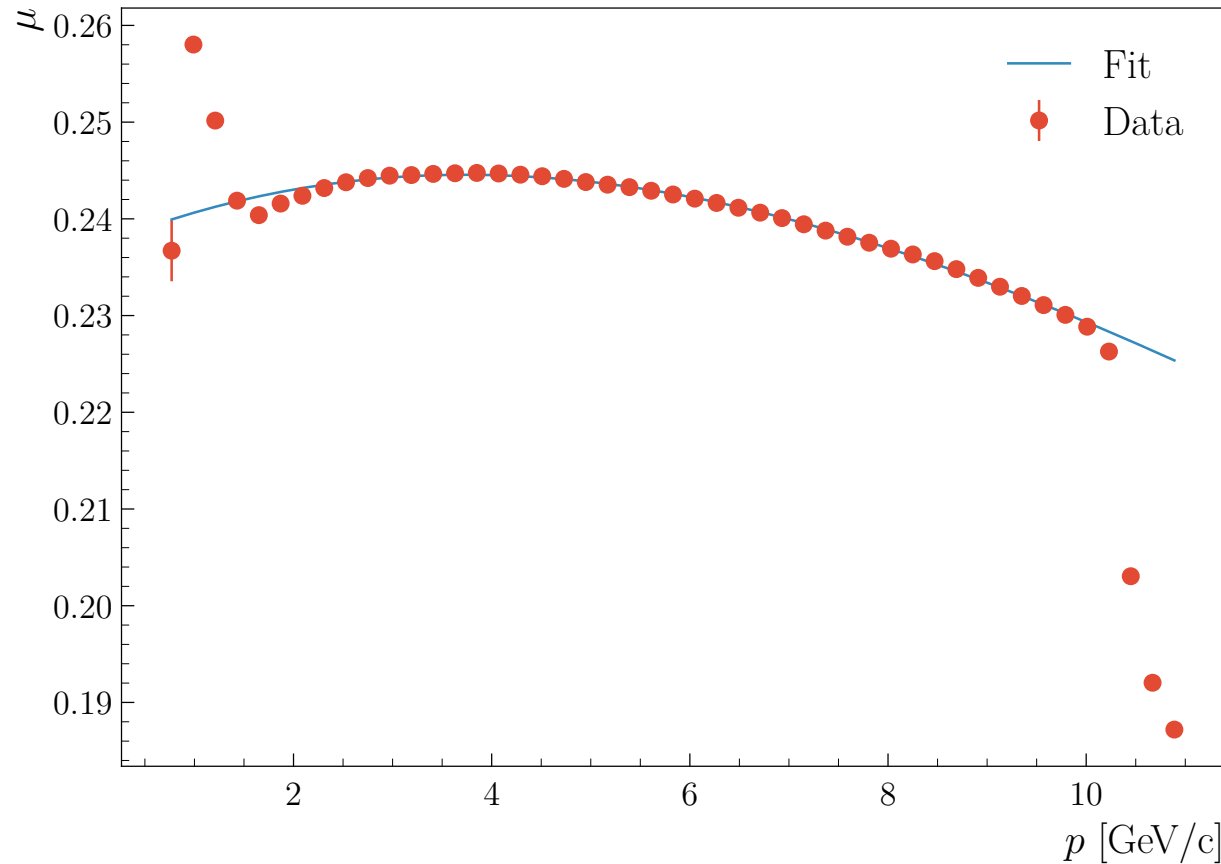
RG-D: Sampling fraction vs momentum cut

- Slice the 2D distribution along the x axis
- Fit the 1D distribution with gaussian
- Extract the mean and sigma, fit with a 3rd order polynomial function
- Cut at $\mu \pm 3\sigma$



RG-D: Sampling fraction vs momentum cut

OBCxC pass0v11 - Electron ($\text{pid} = 11, -4000 < \text{status} \leq -2000, |\text{chi2pid}| < 5$)

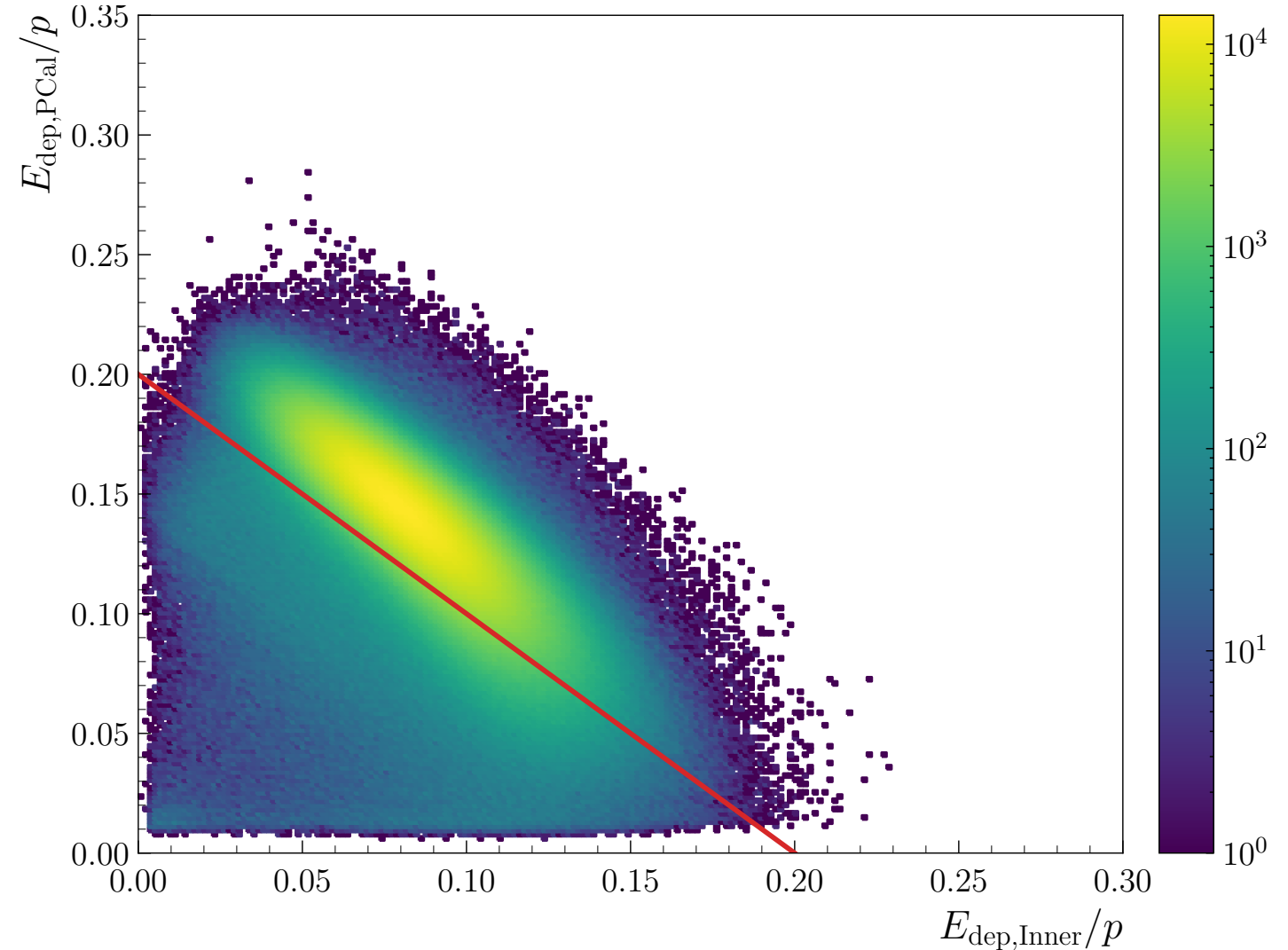


RG-D: Inner calorimeter and the PCal SF cut

- If momentum of the electron is greater than 4.5 GeV:

$$\frac{E_{\text{dep,Inner}}}{p} < 0.2 - \frac{E_{\text{dep,PCal}}}{p}$$

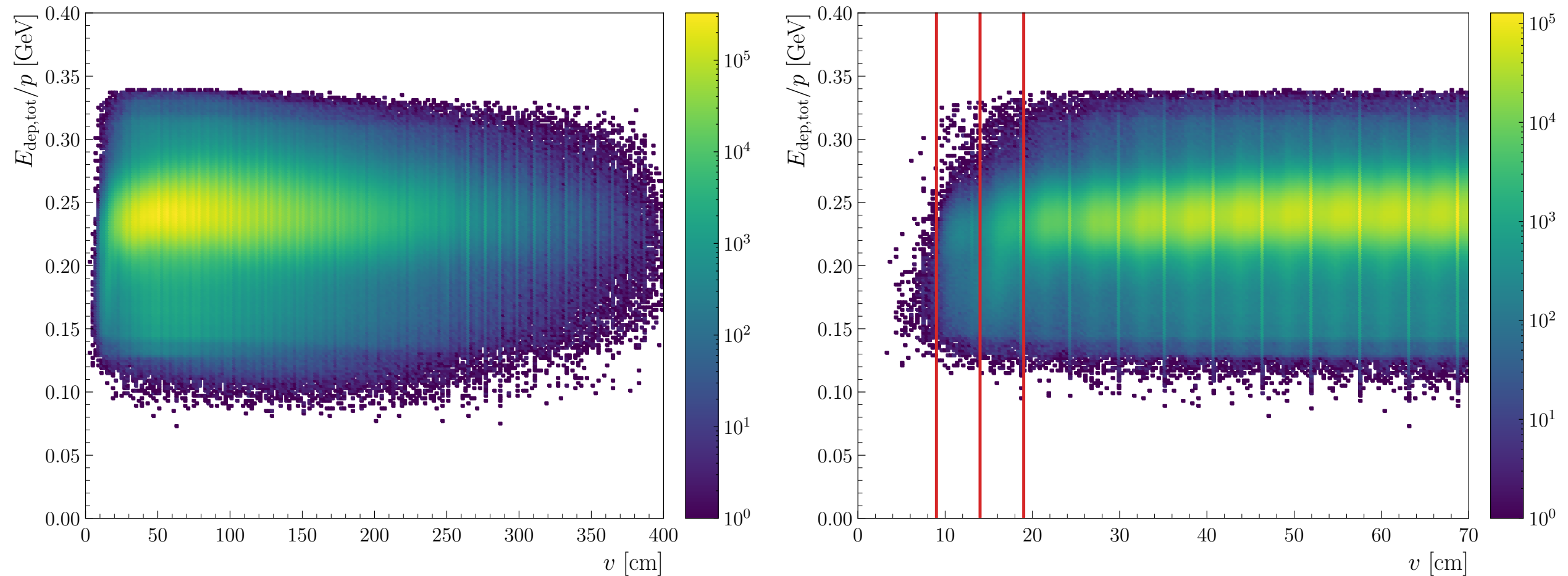
OBCxC pass0v11 - Electron ($\text{pid} = 11, -4000 < \text{status} \leq -2000, |\text{chi2pid}| < 5$)



RG-D: v and w cuts on the PCal

- To remove the edge of the calorimeter, we can cut on the v and w:
 - v is the distance on the V-side, and w is the distance on the W-side

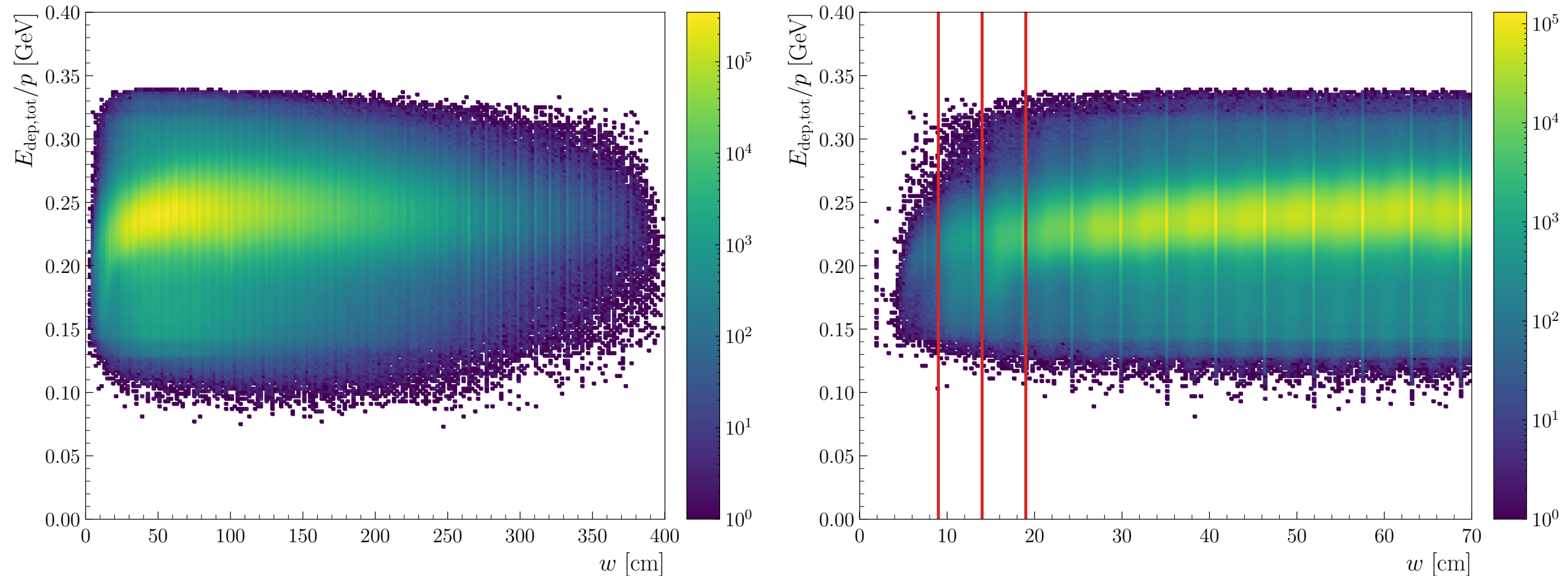
OBCxC pass0v11 - Electron ($\text{pid} = 11, -4000 < \text{status} \leq -2000, |\text{chi2pid}| < 5$) - All sectors



RG-D: v and w cuts on the PCal

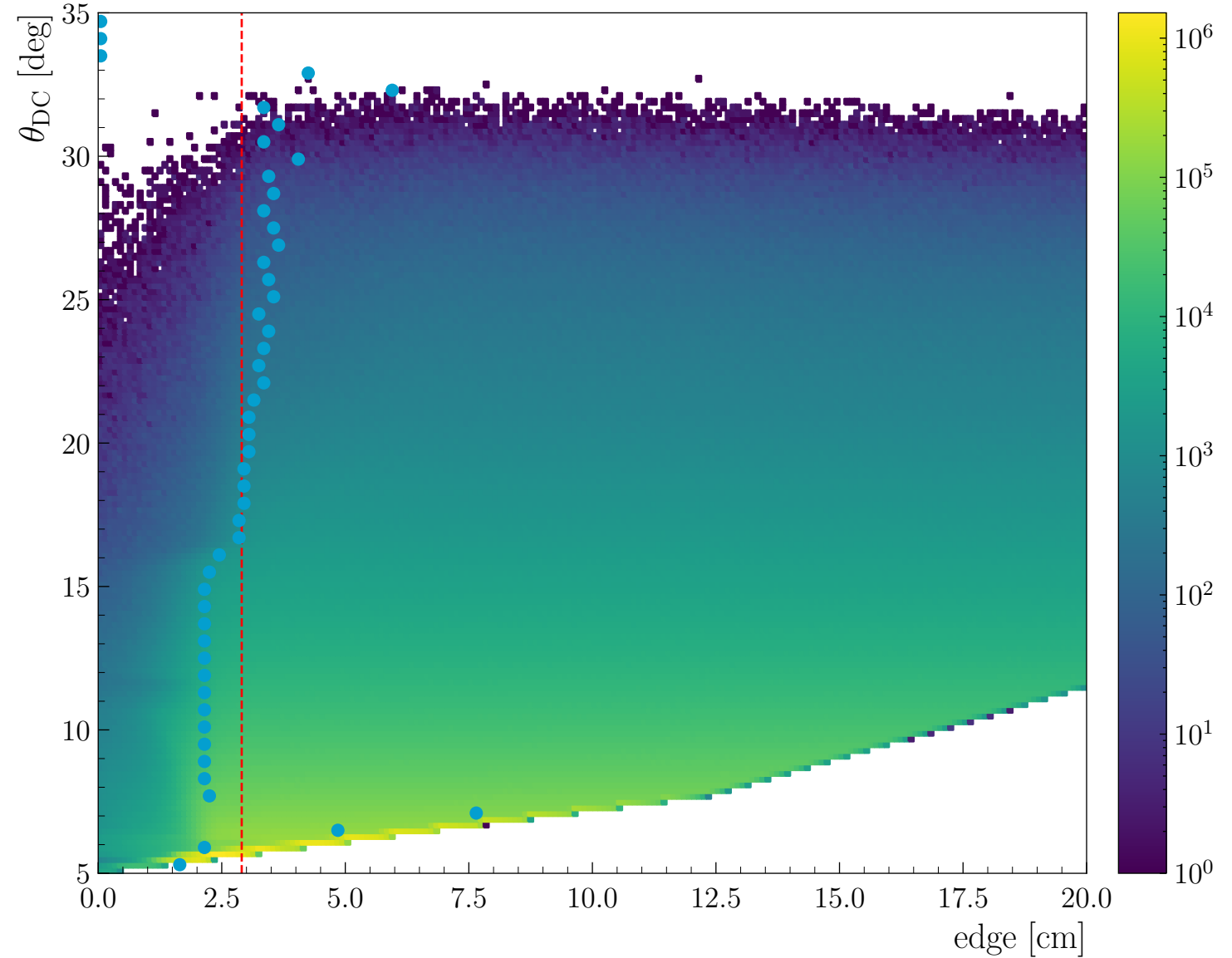
- To remove the edge of the calorimeter, we can cut on the v and w:
 - v is the distance on the V-side, and w is the distance on the W-side

OBCxC pass0v11 - Electron ($\text{pid} = 11, -4000 < \text{status} \leq -2000, |\text{chi2pid}| < 5$) - All sectors



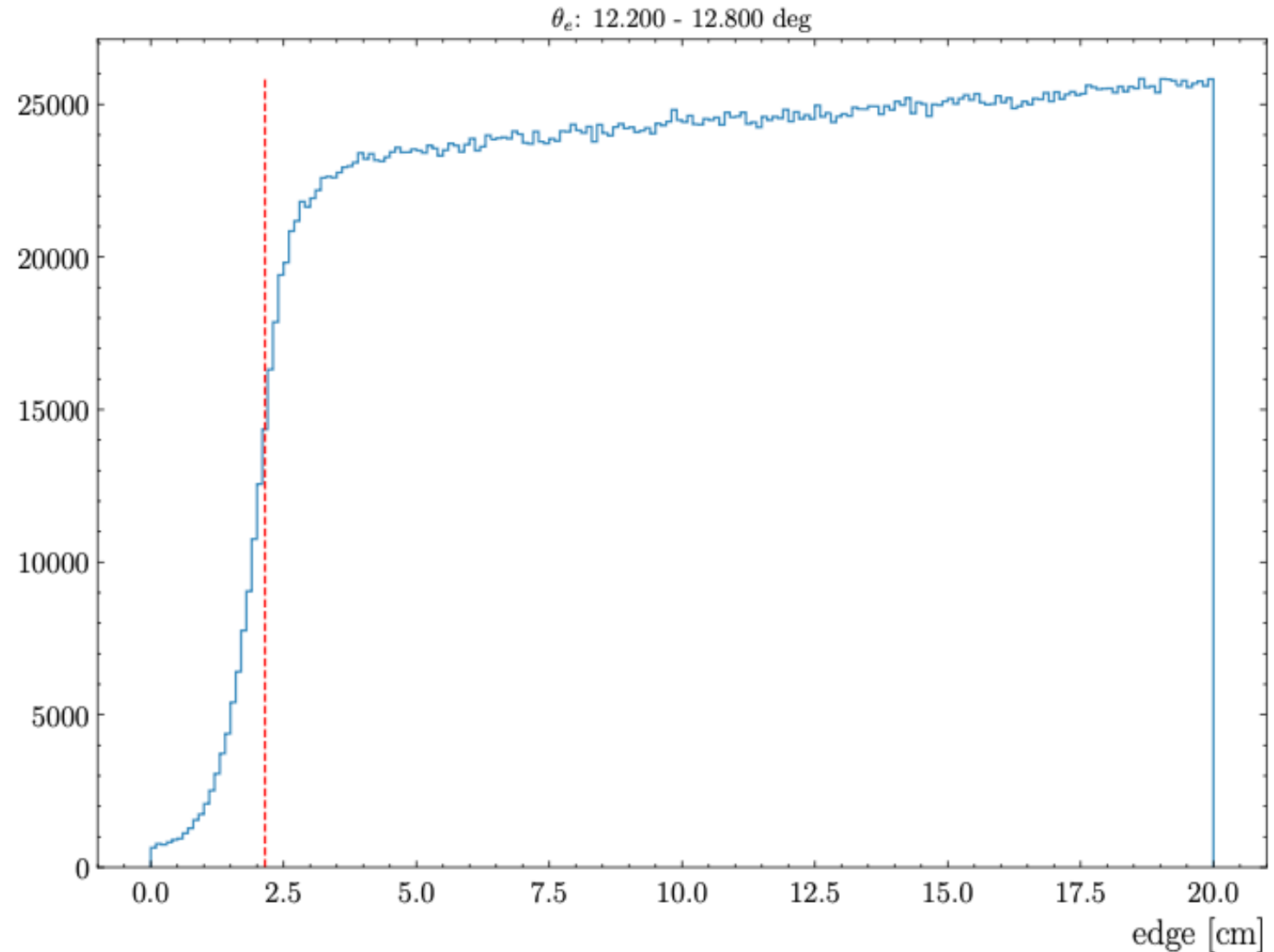
RG-D: DC fiducials cuts using the edge

- Slice the 2D distribution along the y axis
- Blue dots are the thresholds:
 - When the bin content is 50% of the maximum for that histogram



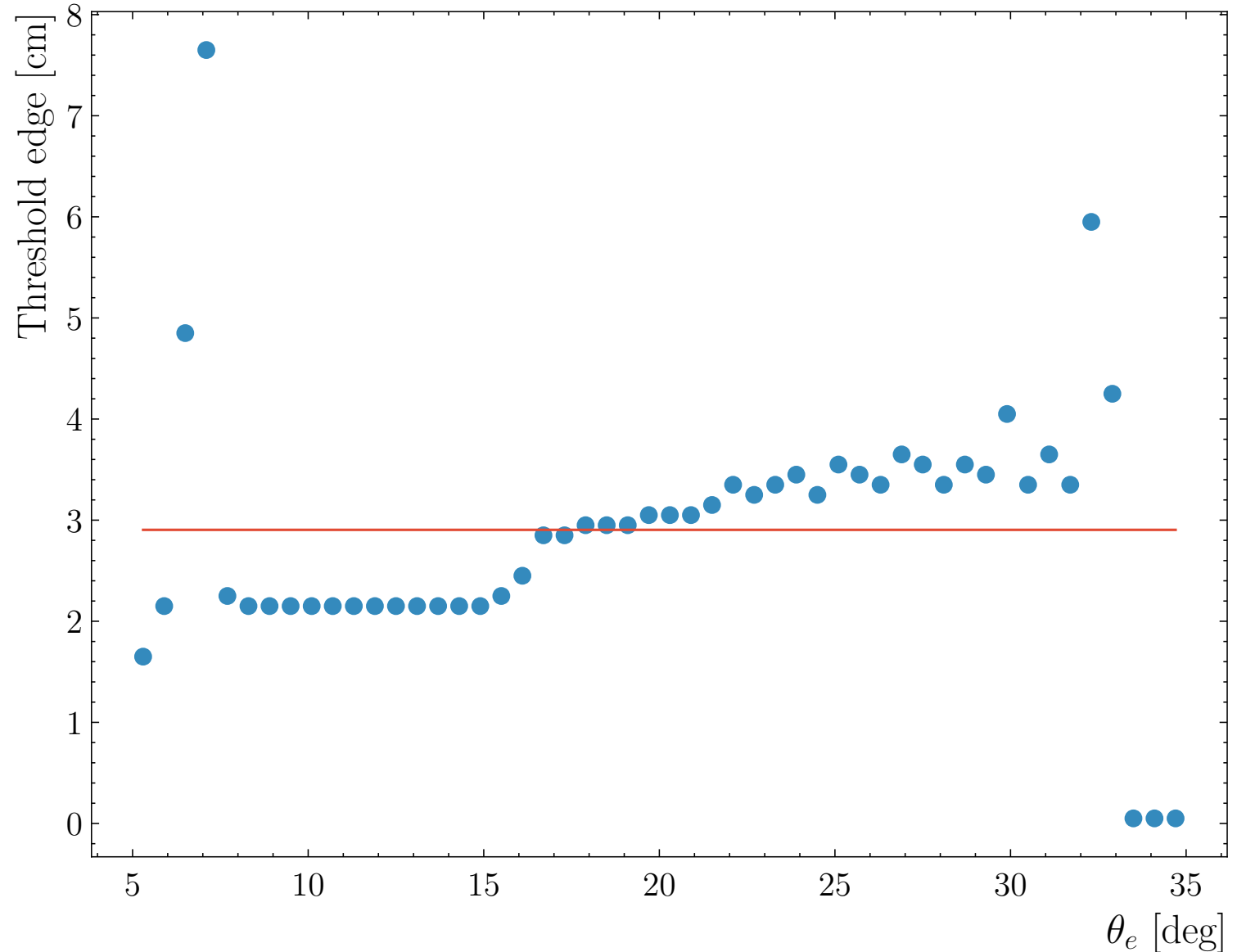
RG-D: DC fiducials cuts using the edge

- Slice the 2D distribution along the y axis
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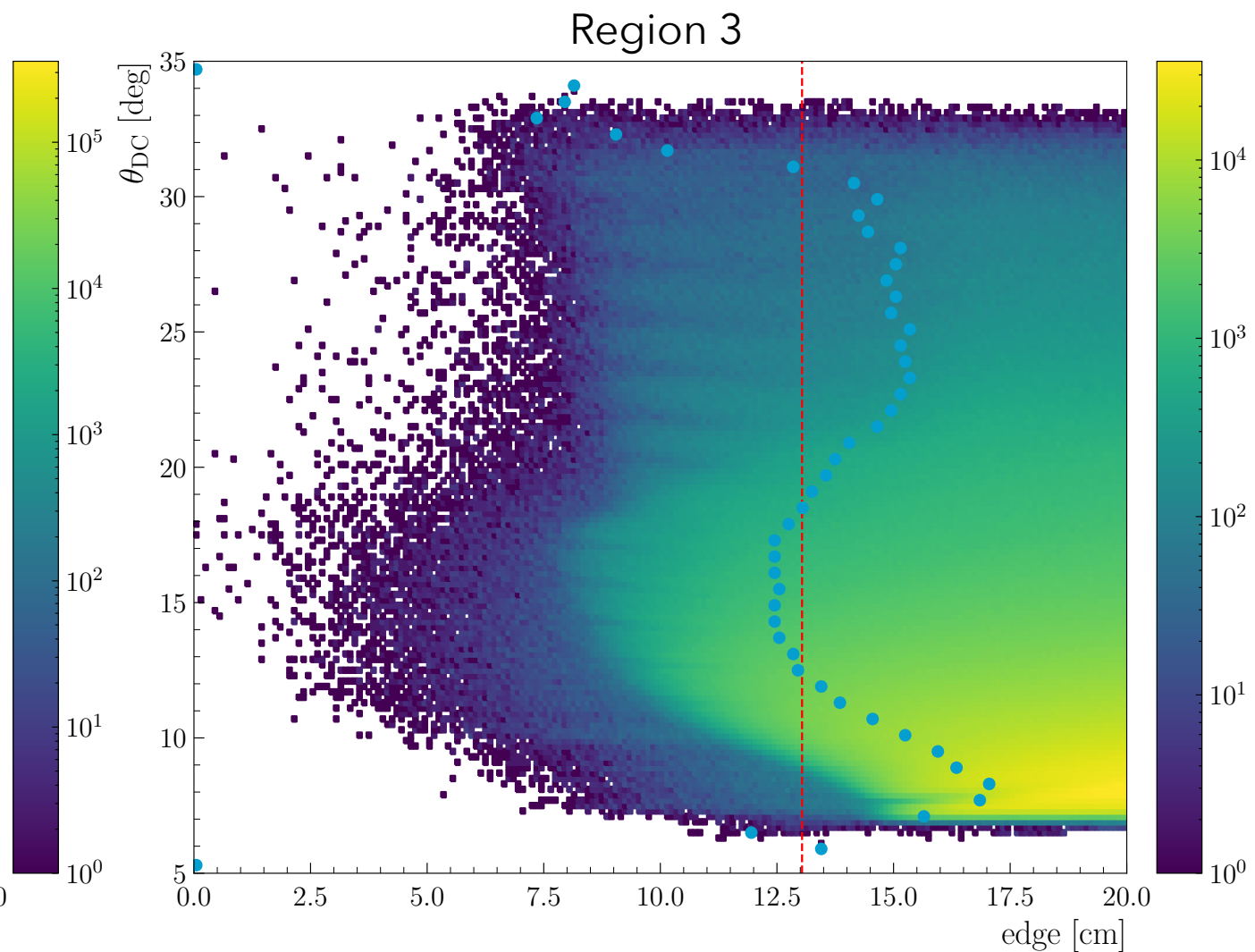
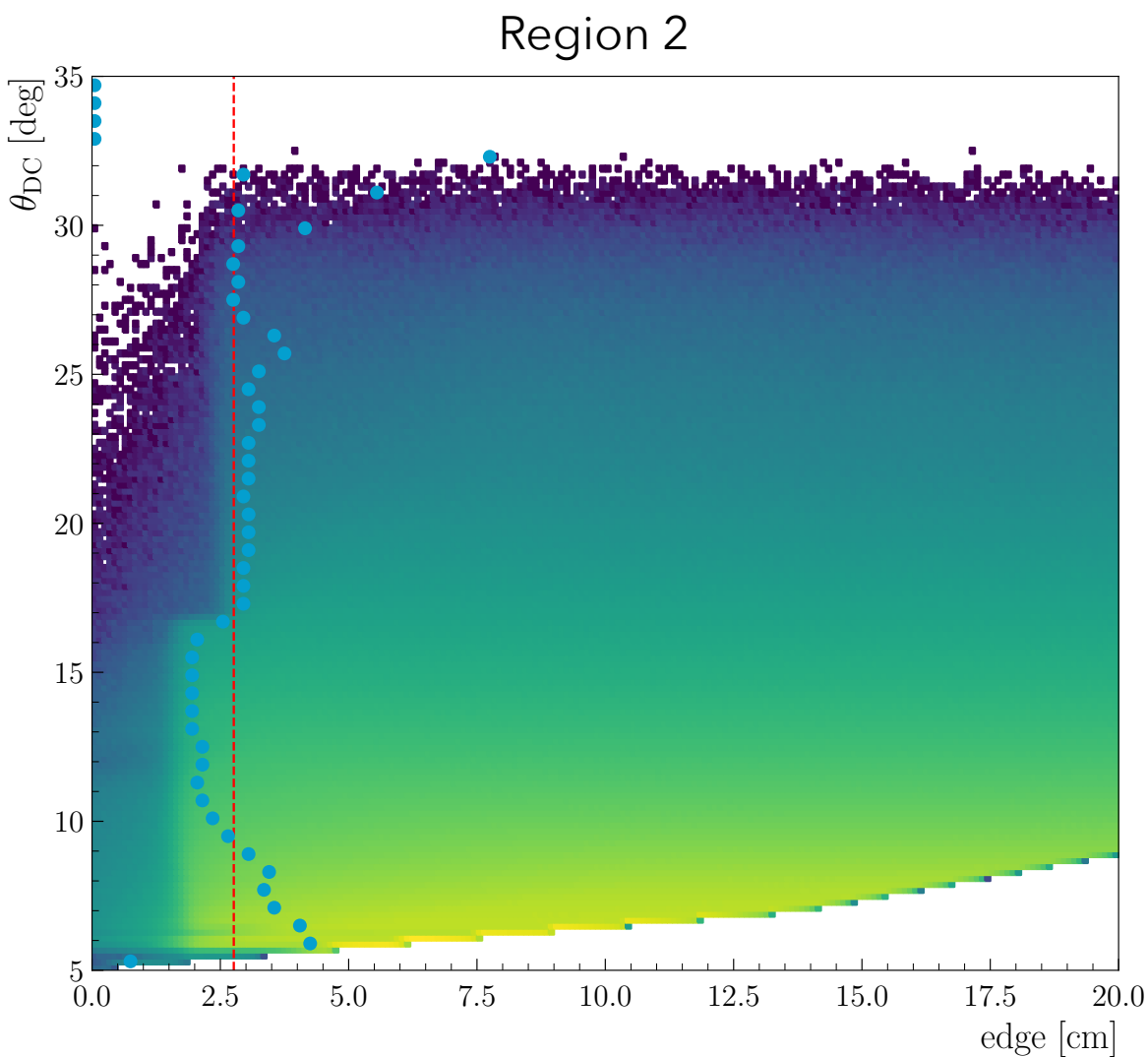
RG-D: DC fiducials cuts using the edge

- Slice the 2D distribution along the y axis
- Blue dots are the thresholds:
 - When the bin content is 50% of the maximum for that histogram
- Fit the threshold with a constant



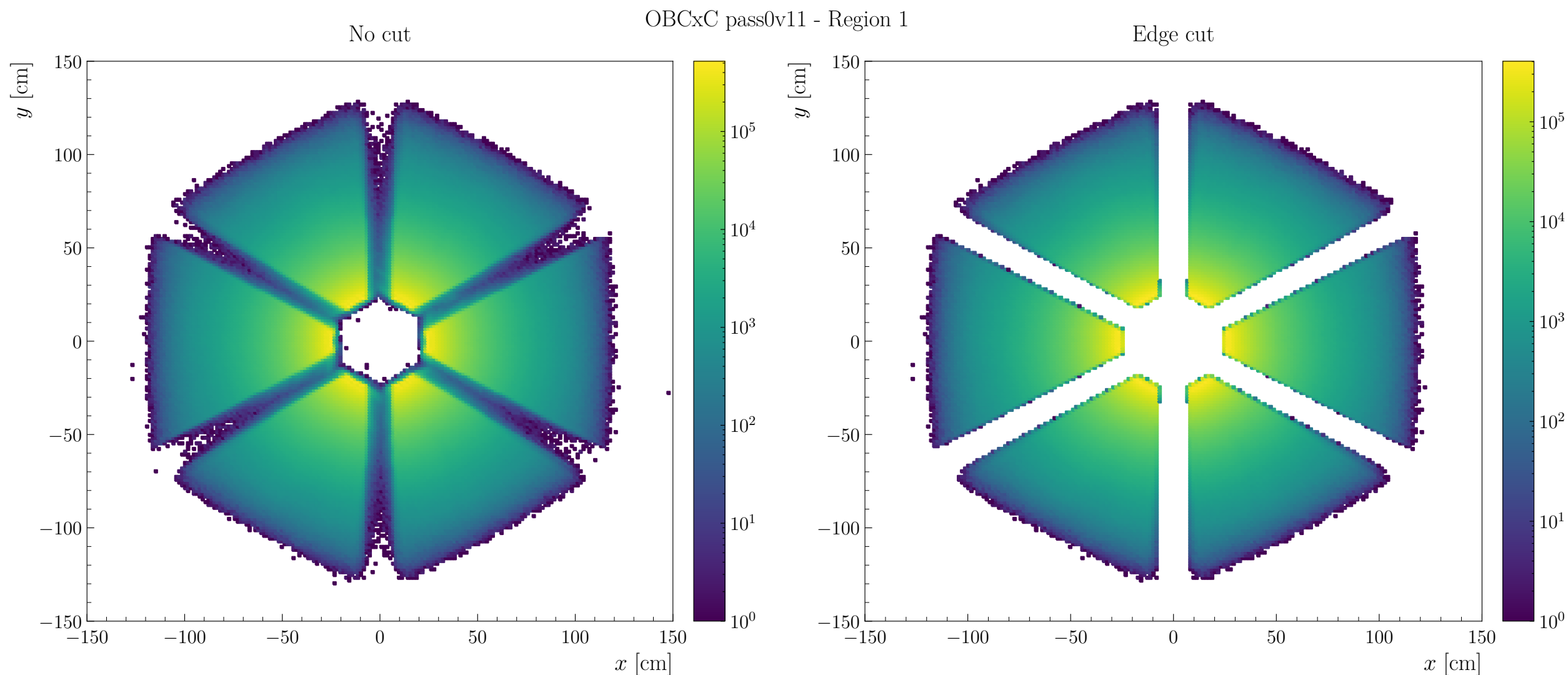
RG-D: DC fiducials cuts using the edge

- Do the same thing for regions 2 and 3



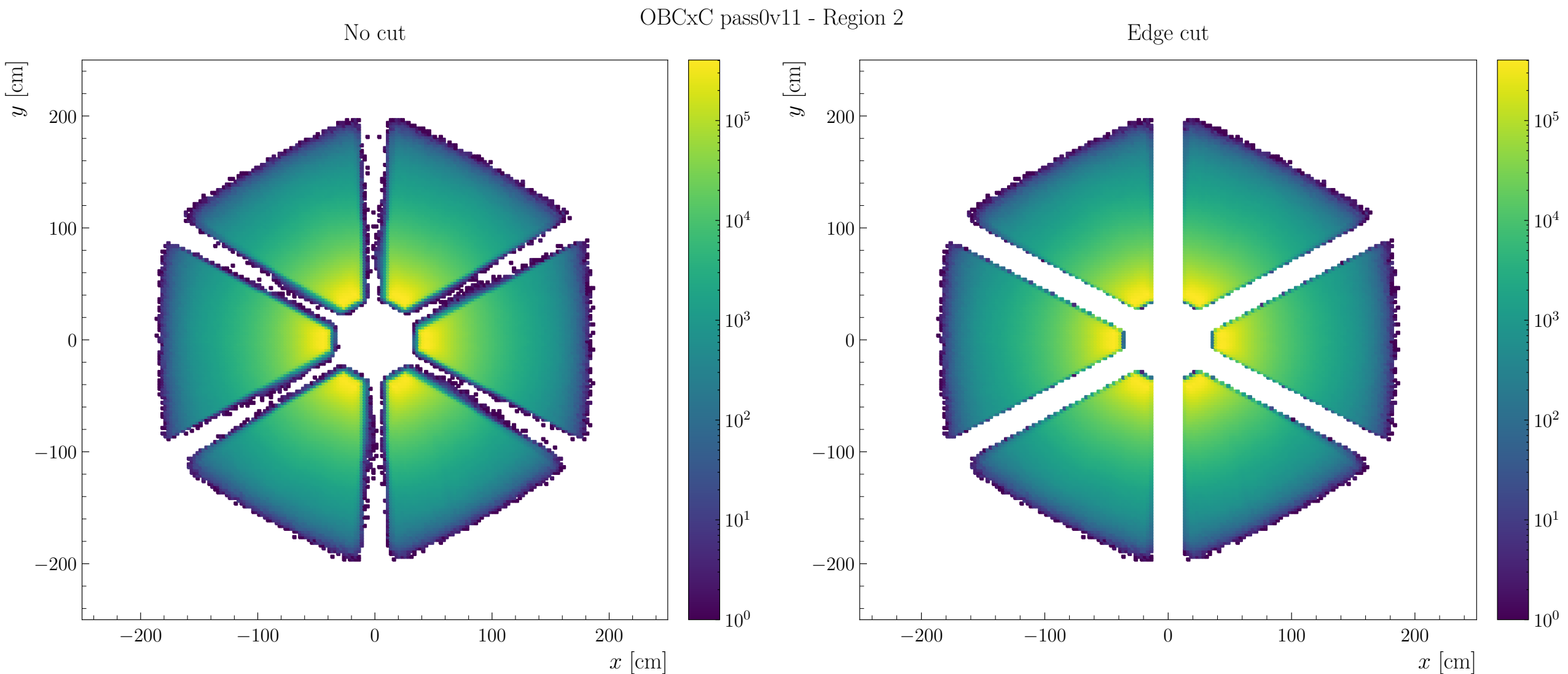
RG-D: DC fiducials cuts using the edge

- Effect of these cuts on x_{DC} and y_{DC} :



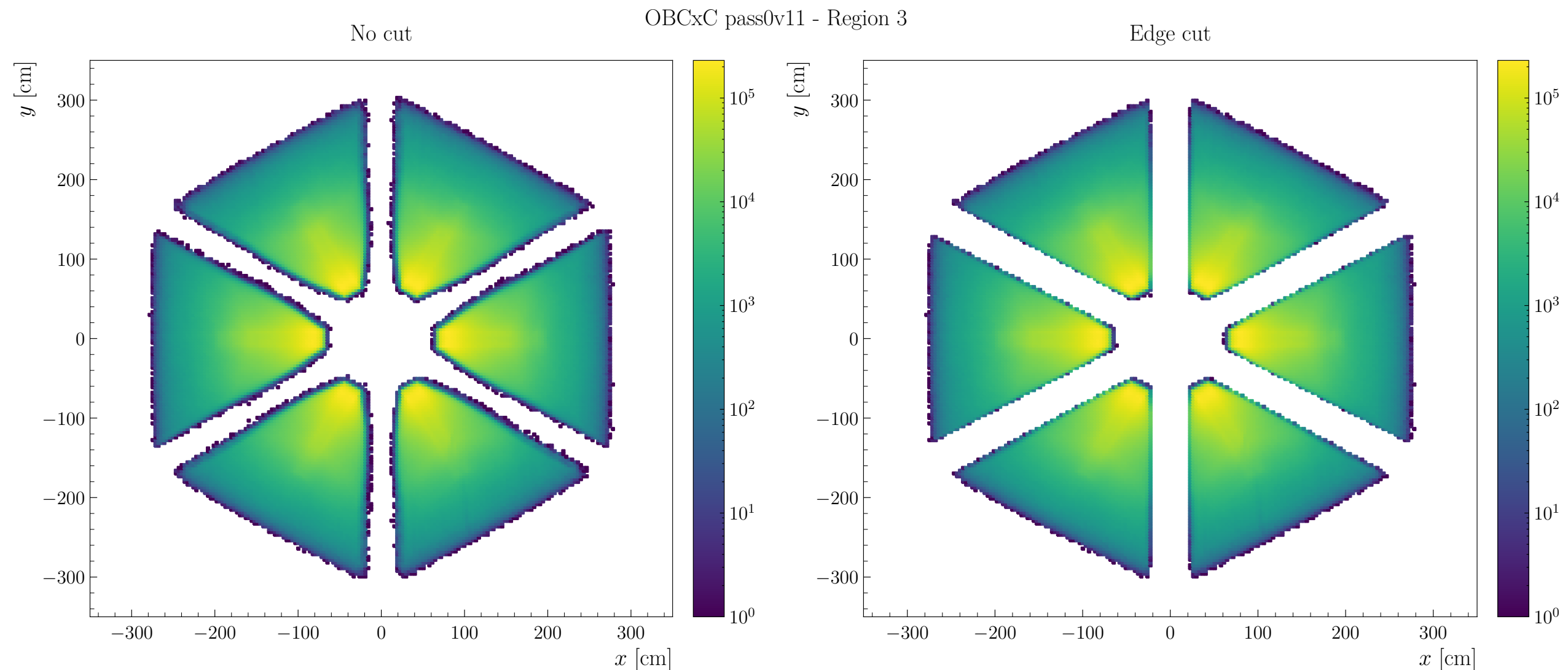
RG-D: DC fiducials cuts using the edge

- Effect of these cuts on x_{DC} and y_{DC} :



RG-D: DC fiducials cuts using the edge

- Effect of these cuts on x_{DC} and y_{DC} :



RG-D: Fiducial cuts effect

- Number of events with electron: 231493276, ratio: 1
- Number of events with EPCal: 231374412, ratio: 0.99
- Number of events with EPCal and SF: 148937729, ratio: 0.64
- Number of events with EPCal, SF and EIn vs EPcal: 139516374, ratio: 0.60
- Number of events with EPCal, SF, EIn vs EPcal and edges: 118785440, ratio: 0.51
- Number of events with EPCal, SF, EIn vs EPcal, edges and v/w: 118785431, ratio: 0.51

RG-D: Vz study

- Try to improve the fit of electron Vz
- Use Double-Sided Crystal Ball function:

$$f(m; m_0, \sigma, \alpha_L, n_L, \alpha_R, n_R) = \begin{cases} A_L \cdot (B_L - \frac{m-m_0}{\sigma_L})^{-n_L}, & \text{for } \frac{m-m_0}{\sigma_L} < -\alpha_L \\ \exp\left(-\frac{1}{2} \cdot \left[\frac{m-m_0}{\sigma_L}\right]^2\right), & \text{for } \frac{m-m_0}{\sigma_L} \leq 0 \\ \exp\left(-\frac{1}{2} \cdot \left[\frac{m-m_0}{\sigma_R}\right]^2\right), & \text{for } \frac{m-m_0}{\sigma_R} \leq \alpha_R \\ A_R \cdot (B_R + \frac{m-m_0}{\sigma_R})^{-n_R}, & \text{otherwise,} \end{cases}$$

- Perform the fit with [RooFit](#) instead of ROOT (it's a package in ROOT)

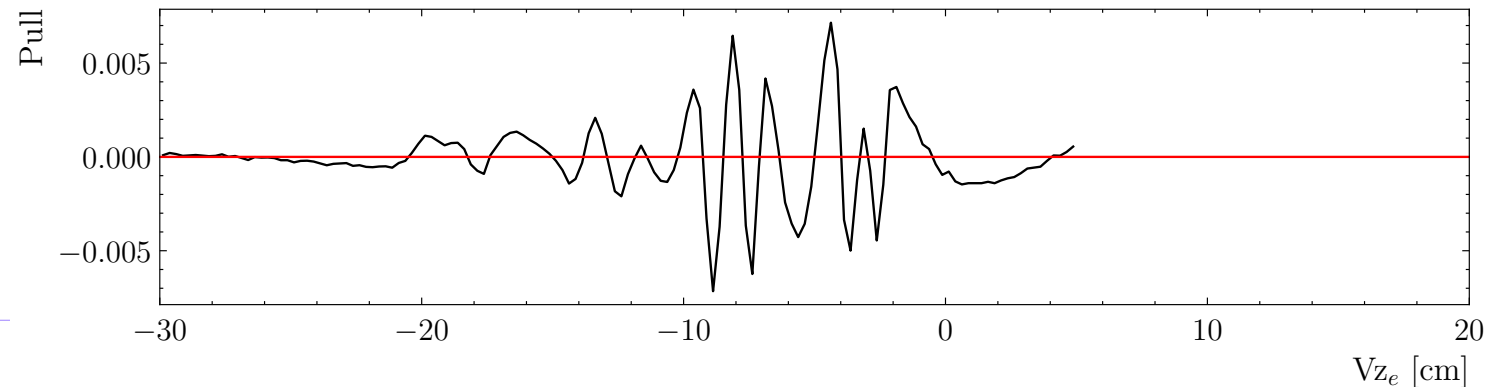
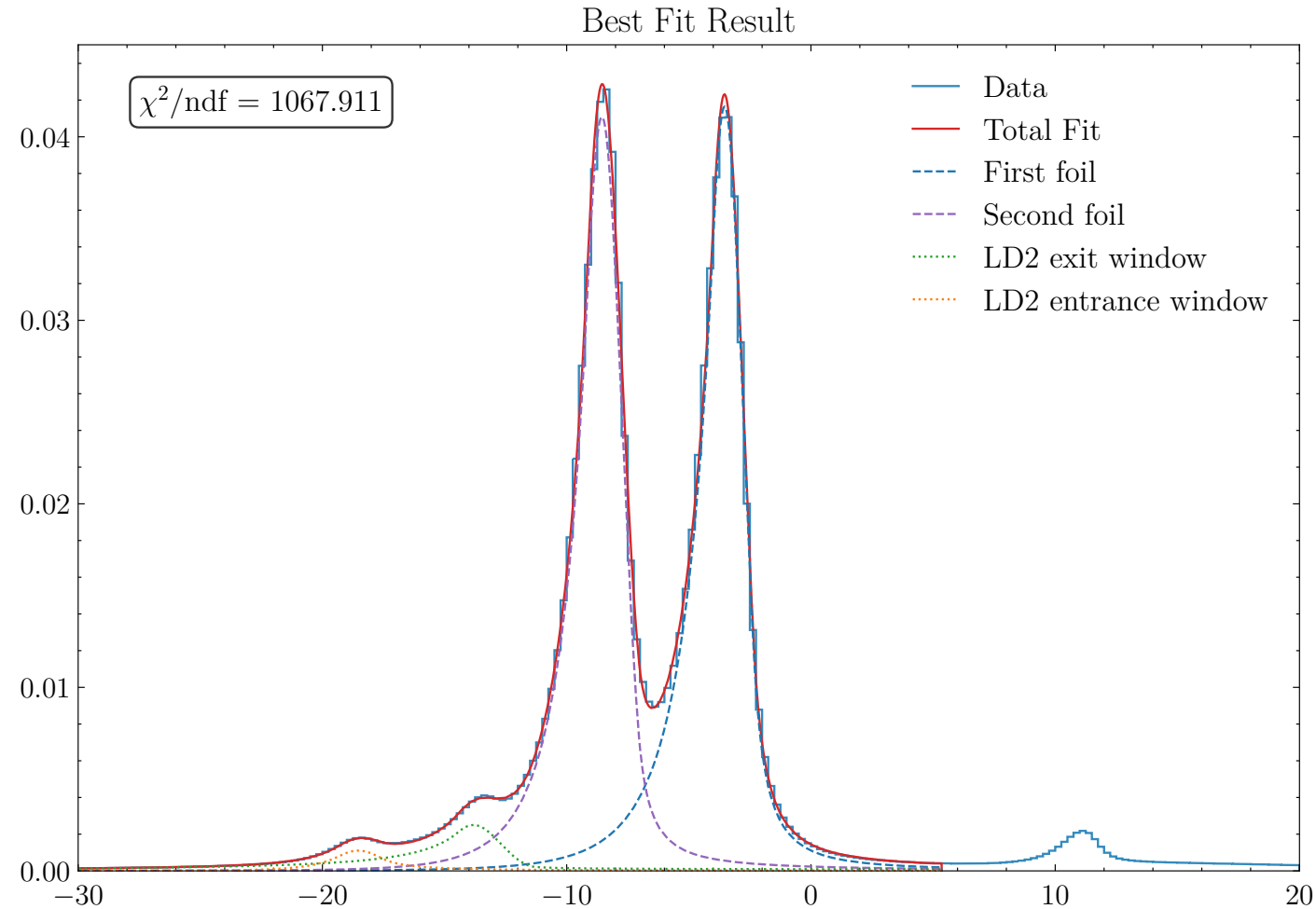
RG-D: Vz study

- Try to improve the fit of electron Vz
- Multi-stage fit:
 - First fit the peak with individual gaussian to get a starting point for the mean and sigma.
 - Fit with 4 DSCBs function 300 times, and randomly adding a small quantity to the values of the parameters each time. Keep the one with the best chi2.
 - Fit with 4 DSCBs function 300 times, but the random step is smaller to slowly improve the fit.
 - Keep the fit with the best chi2.
- We can use the same fit function and same initials parameters for all targets and inbending and outbending

RG-D: Vz study

- Electron, carbon target
- Cuts:
 - Pid=11
 - Status < 0
 - 2000 <= |status| < 4000
 - |chi2pid| < 5

$$\text{Pull} = \frac{x_{\text{data}} - x_{\text{fit}}}{\sigma_{\text{data}}}$$



RG-D: Vz study

- Electron, carbon target

- Cuts:

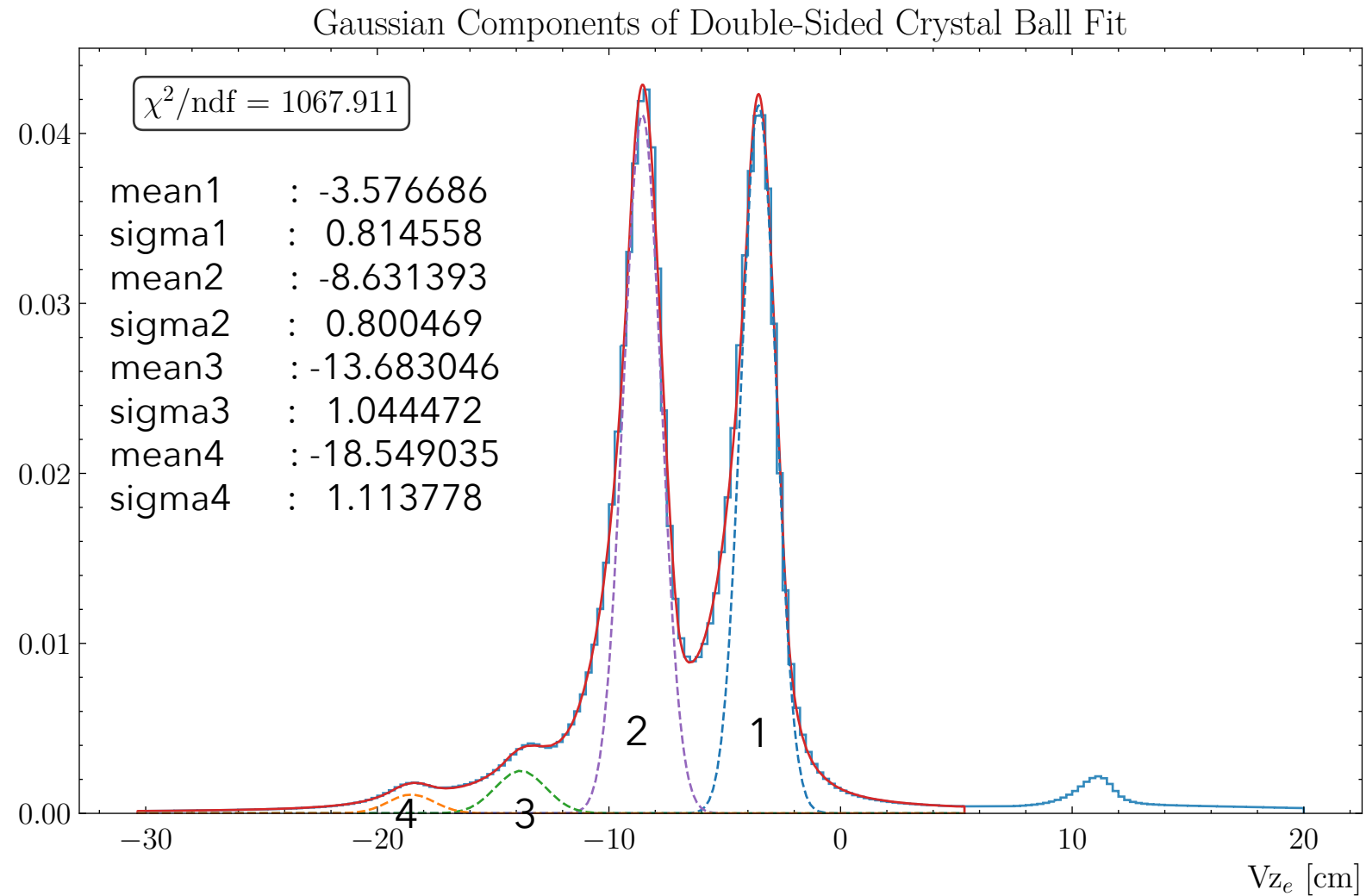
- $P_{id}=11$

- $Status < 0$

- $2000 \leq |status| < 4000$

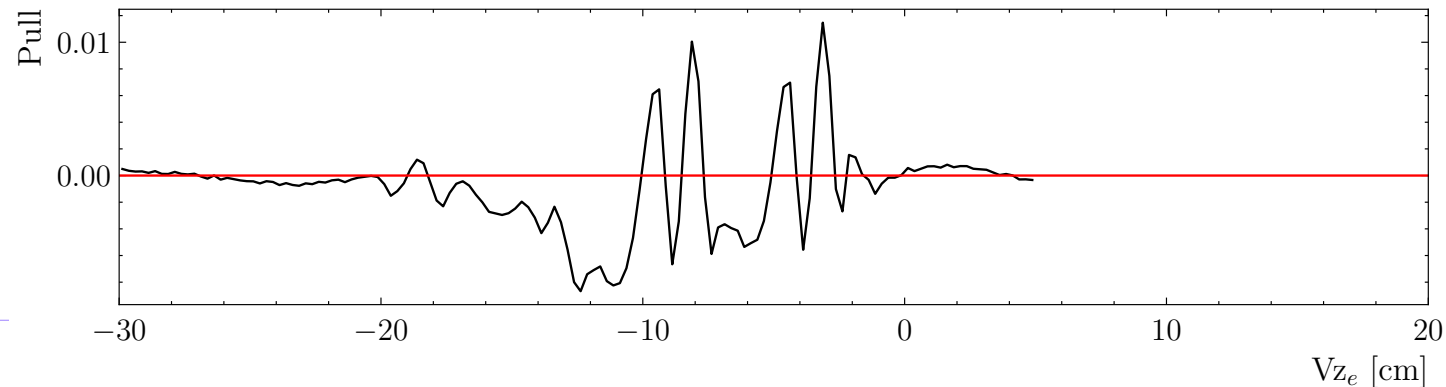
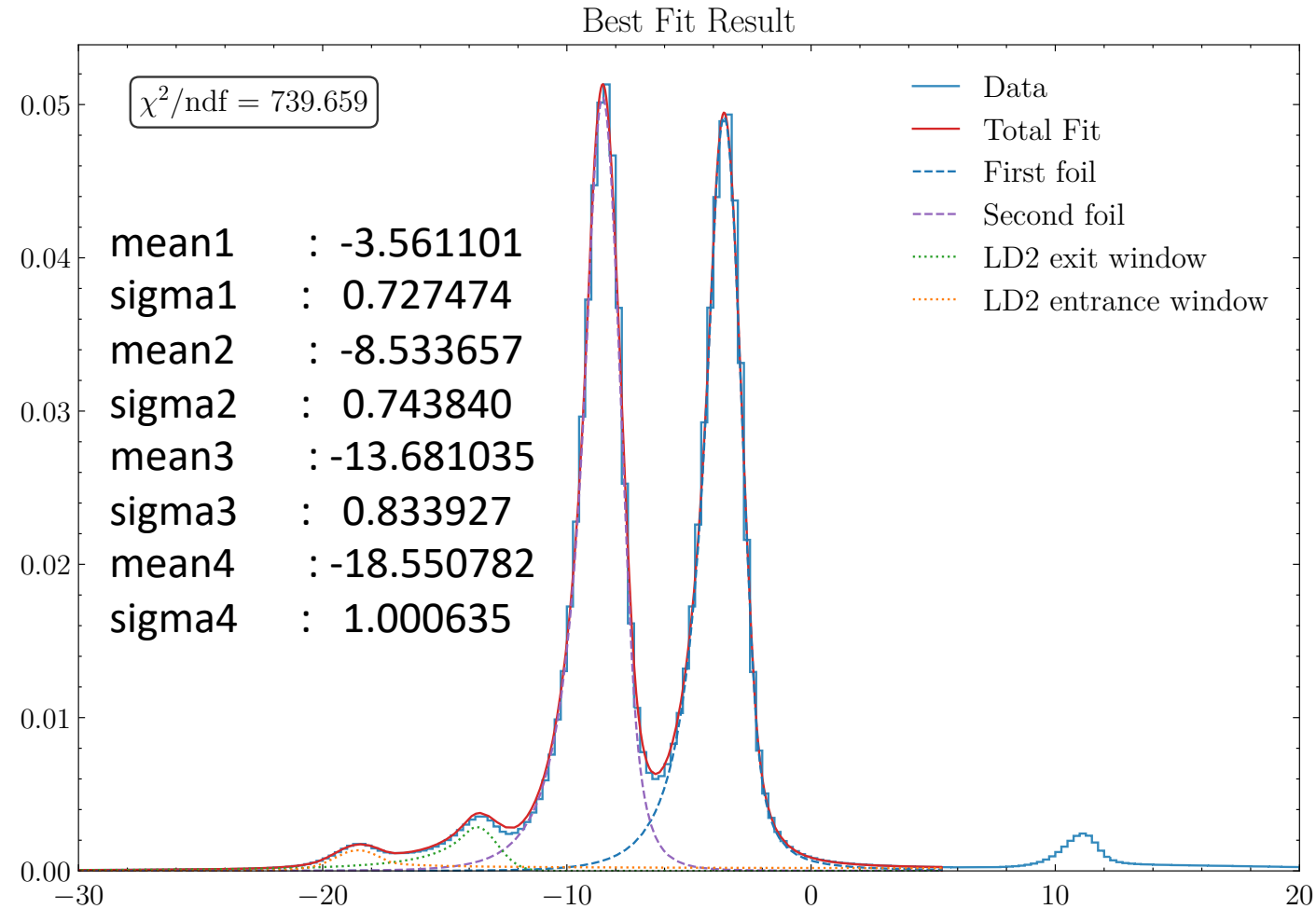
- $|chi2pid| < 5$

- We can extract the gaussian part of the DSCB



RG-D: Vz study

- Electron, carbon target
- Cuts:
 - Pid=11
 - Status < 0
 - $2000 \leq |\text{status}| < 4000$
 - $|\text{chi2pid}| < 5$
 - SF vs p: 3 sigma
 - $E_{\text{pcal}} < 0.07 \text{ GeV}$
 - $E_{\text{in}}/p < 0.2 - E_{\text{Pcal}}/p$ if $p > 4.5 \text{ GeV}$
 - $u, w > 9 \text{ cm}$
 - edge_{DC} cuts



Some stuffs

- I'll be the cooking chef for ALERT beginning this Wednesday.
- Regarding the CLAS collaboration meeting: Should I present the update for RG-D?

To do list

Task	expected	done	comment
Improve pid, fiducials cuts	04/21		Work on the DC fiducials, need to produce a PDF with all the plots and details, some issues with defining the DC limit. look good now
Produce physics plots for pass1 review	04/30	done	
Finish the event mixing method	05/04		
Extract the multiplicity ratio, pT	05/15		
improve code that generated all track candidates (ALERT)	05/15		
Test the AI on ^4He	06/02		
Write the AI study on the wiki RG-D	05/12	done	Need to write the background merging part